

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**ARTIFICIAL INTELLIGENCE (22AD42) – Question Bank****Unit - 1**

1. Discuss in detail the History of AI -8M
2. Outline the Winograd Schema with an example -7M
3. Write a short notes on Language, mind, philosophy, reasoning -6M
4. Define Intelligent agents -3M
5. Explain the steps and features of State Space Search -7M
6. Differentiate between BFS and DFS -5M
7. Illustrate the Heuristic Search Problem with an example -7M
8. Explain Depth first iterative deepening -4M
9. Discuss in detail Hill climbing with an example -8M
10. Explain the Travelling Salesman problem with an example -7M

Unit -2

11. Outline the concept of Escaping Local Optima -3M
12. Discuss about the Random Walk Algorithm -5M
13. Infer about the Simulated annealing with an algorithm -7M
14. Infer about the Population based methods -3M
15. Illustrate the Genetic Algorithms with an example -8M
16. Explain the Eight Queen's problem -7M
17. Illustrate the concept of Ant colony optimization -8M
18. List the applications of TSP -3M
19. Discuss in detail Branch and Bound -7M
20. Interpret the A* algorithm with an example -8M
21. Explain the informed Heuristic functions -4M
22. Discuss the Emergent systems -4M

Unit – 3

23. Discuss the Results related to Admissibility of A* -8M
24. Explain the Iterative Deepening A* -7M
25. Discuss the advantages and Disadvantages of IDA -4M
26. Illustrate the Board Games and Game Trees in Game playing -8M
27. Interpret the Algorithm MiniMax with an example -8M
28. Infer the Algorithm AlphaBeta along with an example -7M

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| 29. Discuss in detail the examples of Go and Backgammon | -8M |
| 30. Explain the types of Game Playing | -4M |
| 31. Outline the applications of search algorithm in AI | -2M |